Perfect maps reflect compactness
Munkres, 26.12

Theorem 1. *A continuous surjection $f : X \to Y$ is perfect if it is closed and every fiber $f^{-1}(y)$ is compact. If f is perfect and Y is compact, then X is compact.*

Proof. We use the characterization of compactness by finite subcovers. Let $\{U_i\}_{i\in\iota}$ be an open cover of X; thus each U_i is open in X and

$$X \subseteq \bigcup_{i\in\iota} U_i.$$

We shall find a finite subset of ι whose corresponding sets still cover X.

For each $y \in Y$, the fiber $f^{-1}(\{y\})$ is compact by hypothesis. It is covered by the open sets U_i, so compactness of the fiber gives a finite set $t(y) \subseteq \iota$ such that

$$f^{-1}(\{y\}) \subseteq \bigcup_{i\in t(y)} U_i.$$

Choose such a finite set $t(y)$ for every $y \in Y$.

For $y \in Y$, put

$$W_y = \bigcup_{i\in t(y)} U_i, \qquad V_y = \big(f(W_y^c)\big)^c,$$

where ^{c} denotes complement. The set W_y is open, being a union of open sets. Hence W_y^c is closed. Since f is a closed map, $f(W_y^c)$ is closed, and therefore V_y is open.

We claim that $y \in V_y$. Indeed, otherwise $y \in f(W_y^c)$, so there would exist $x \in W_y^c$ with $f(x) = y$. Then $x \in f^{-1}(\{y\})$, and hence by the choice of $t(y)$ we would have $x \in W_y$, a contradiction. Thus

$$Y \subseteq \bigcup_{y\in Y} V_y,$$

so $\{V_y\}_{y\in Y}$ is an open cover of Y.

Since Y is compact, there is a finite set $S \subseteq Y$ such that

$$Y \subseteq \bigcup_{y\in S} V_y.$$

We claim that the finite set of indices

$$\bigcup_{y\in S} t(y)$$

gives the required finite subcover of X. Let $x \in X$. Since $f(x) \in Y$, there exists $y \in S$ such that $f(x) \in V_y$. Then $x \in W_y$: otherwise $x \in W_y^c$, which would

imply $f(x) \in f(W_y^c)$, contradicting $f(x) \in V_y = \left(f(W_y^c)\right)^c$. Hence there is some $i \in t(y)$ with $x \in U_i$. Since $y \in S$, we have $i \in \bigcup_{y \in S} t(y)$. Therefore

$$X \subseteq \bigcup_{i \in \bigcup_{y \in S} t(y)} U_i.$$

This is a finite subcover, since $\bigcup_{y \in S} t(y)$ is a finite union of finite sets. Thus every open cover of X has a finite subcover, so X is compact. $\square$